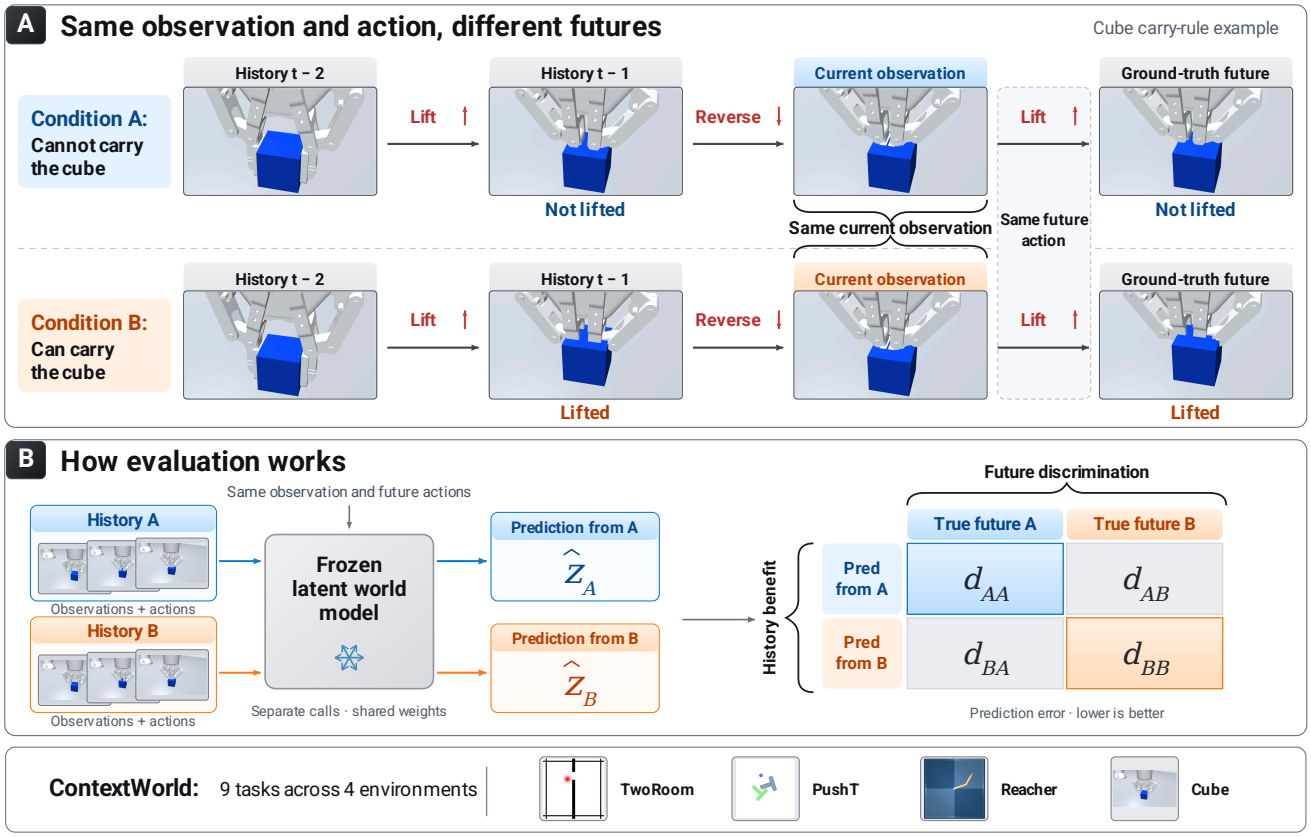



**Figure 1. Controlled prediction and history comparisons.** A: Cube carry-rule example, re-rendered for clarity. A lift command and its reverse reveal whether the cube is carried, then return both conditions to the same current observation. Repeating the lift produces different futures. Commands label transitions. B: The frozen model receives observations and actions and produces latent predictions. Rows compare futures for a fixed prediction; columns compare histories for a fixed target.

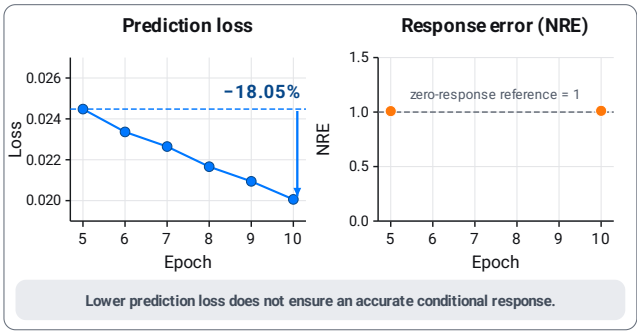



**Figure 4. Prediction loss and conditional response.** PLDM on contact friction, Original → Mixed (Joint). Recorded prediction loss falls by 18.05% from epochs 5 to 10, while training-query NRE stays near the zero-response reference of 1. Loss uses mixed training batches; NRE is measured on fixed paired training queries at the two endpoints. Full task and scheme comparisons remain in the manuscript appendix.

### Prediction Loss and Adaptation

Weak evaluation performance may reflect poor generalization or conditional responses that were not learned well during training. We compare prediction loss with NRE on fixed training queries at epochs 5 and 10.

On friction, PLDM joint training reduces prediction loss by 18.05%, yet training-query NRE remains near the zero-response reference. On damping with the encoder frozen, loss approaches convergence while NRE improves but remains high. Changing the scheme helps: freezing yields a stronger conditional response on friction, whereas joint training performs better on damping.

These results expose a disconnect between prediction-loss optimization and history-conditioned adaptation, even on training data. Prediction loss aggregates errors across mixed data, whereas NRE directly measures error in the predicted differences between conditions.